Machine learning models in genome wide association studies of complex traits

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INTRODUCTION

Machine Learning (ML) applied to health sciences is one of the domains which mostly interested researchers in the last decade. Genome-wide association studies investigate the individual association of many genetic loci (several hundred thousand) with complex traits. Thus, ML may find a possible application in these studies to unravel multi-locus genetic patterns associated with the trait variability. Here, we applied some ML approaches on simulated data to assess the ability of pattern recognition algorithms (Machine learning models) to accurately select relevant features (SNPs) underlining the aetiopathogenesis of the disorder and (2) how the presence of a huge number of noisy features impacts on the selection performance of truly associated signals.

MATERIAL AND METHODS

- R package ***phenotypeSimulator***, simulates a genotype data of **200'000 independant SNPs** for **1000 individuals**.These genotype data have been used as input of ***SeqSIMLA2*** to generate a prevalence disease (**prev = 0.13**) genome-phenome case-control data composed of **200'000 SNPs and 1'200 samples** (650 controls, 550 cases). Where **snp56265, snp61523, snp62831, snp70478** have been set as causals with respectively an arbitrary **Odds ratio of 4,5,3,2** and the **Minor Allele frequency at 0.395, 0.104, 0.404,0.186**.
- *Plink*; association study and the selection of three subsets based on the pvalue (**Pv <= 0.01, Pv <= 0.05, Pv <= 0.1**).
- 2 boostings models ***(CatBoost*** classifier (Cat), **XGBoost** classifier (XG)), **RandomForest** classifier (RF), ***shap***, and ***R chorddiag*** have been used to analyze, produce result and visualize the selected data.

3 subsets:

- **Pv <= 0.01** composed of **15'925 SNPs**
- **Pv <= 0.05** composed of **36'689 SNPs**
- **Pv <= 0.1** composed of **52'916 SNPs**

RESULTS

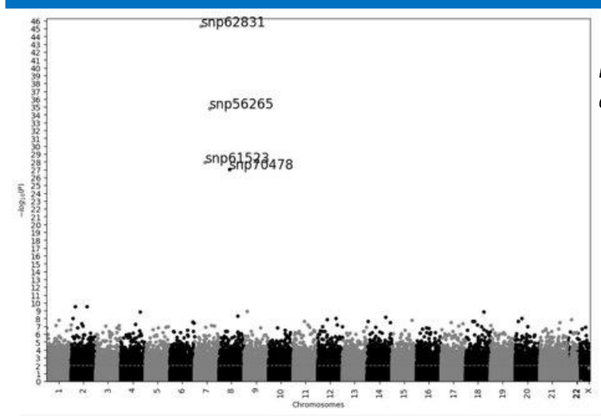

Figure 1. Manhattan plot of association study

Table I. models evaluation scores

	ACCURACY			PRECISION			RECALL		
	Cat	XG	RF	Cat	XG	RF	Cat	XG	RF
Pv_001	0.84	0.84	0.80	0.83	0.85	0.78	0.89	0.84	0.85
Pv_005	0.83	0.85	0.73	0.82	0.87	0.74	0.89	0.86	0.80
Pv_01	0.85	0.85	0.76	0.85	0.87	0.81	0.89	0.89	0.80

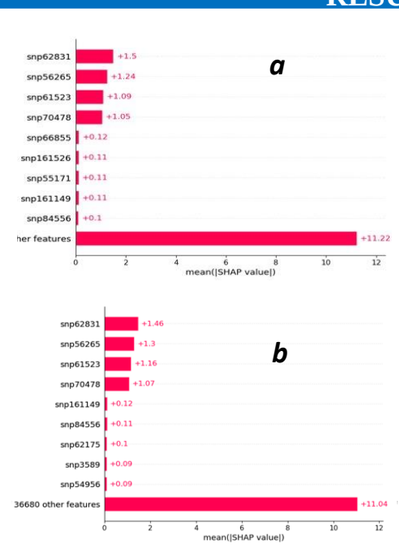

Figure 2. Contribution of SNPs on the models prediction. a. pv= 0.01; a. pv= 0.01; a. pv= 0.01.

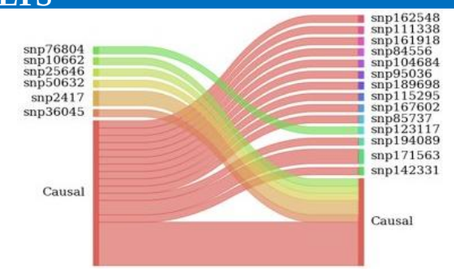

Figure 3. SNPs interactions

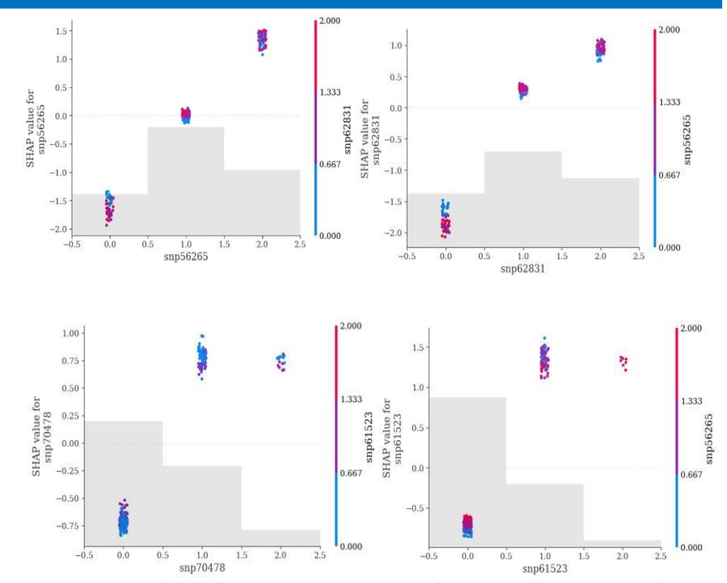

Figure 4. Partial dependance plots of most contributor SNPs

CONCLUSION

All the 2 boosting classifiers showed a good accuracy score (range: 83-85%) for all the datasets. RF performed less with an accuracy of 73-80% . The risk-SNPs (higher contribution) were always selected by all the classifiers and this was true independently from the number of no associated SNPs included into the study (15'925, 36'689, 52'916 SNPs).

In future work, we will simulate and investigate different genetic configurations (lower Odd ratio, several causal SNPs) to study in more details how suitable ML models could be used for genomic purpose.

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